

nvidia / **NV-Embed-v2**   like 515 Follow  NVIDIA 67.4k Feature Extraction  Transformers  Safetensors  sentence-transformers  Englishnvembed mteb custom_code  Eval Results (legacy)  arxiv:2405.17428  arxiv:2407.15831 License: cc-by-nc-4.0 Deploy ▾→ Copy to bucket **NEW**

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Model card

 Files xet

Community 46

Introduction

We present NV-Embed-v2, a generalist embedding model that ranks No.1 on the Massive Text Embedding Benchmark ([MTEB benchmark](#)) (as of Aug 30, 2024) with a score of 72.31 across 56 text embedding tasks. It also holds the No.1 in the retrieval sub-category (a score of 62.65 across 15 tasks) in the leaderboard, which is essential to the development of RAG technology.

NV-Embed-v2 presents several new designs, including having the LLM attend to latent vectors for better pooled embedding output, and demonstrating a two-staged instruction tuning method to enhance the accuracy of both retrieval and non-retrieval tasks. Additionally, NV-Embed-v2 incorporates a novel hard-negative mining methods that take into account the positive relevance score for better false negatives removal.

For more technical details, refer to our paper: [NV-Embed: Improved Techniques for Training LLMs as](#)

Downloads last month

19,724

Safetensors

Model size 8B params

Tensor type F16  Files info

Inference Providers **NEW**

 Feature Extraction

This model isn't deployed by any Inference Provider.



4 Ask for provider support

Model tree for nvidia/NV-Embed-v2

 Finetunes 6 models

Spaces using nvidia/NV-Embed... 32

 mteb/leaderboard dwb2023/hf_extractor ashwiniai/medrag-multi-modal

Model Details

- Base Decoder-only LLM: Mistral-7B-v0.1
- Pooling Type: Latent-Attention
- Embedding Dimension: 4096

How to use

Here is an example of how to encode queries and passages using Huggingface-transformer and Sentence-transformer. Please find the required package version here.

Usage (HuggingFace Transformers)

```
import torch
import torch.nn.functional as F
from transformers import AutoTokenizer, AutoModel

# Each query needs to be accompanied by an instruction
task_name_to_instruct = {"example": "Given a question, answer it."}

query_prefix = "Instruct: "+task_name_to_instruct["example"]
queries = [
    'are judo throws allowed in wrestling?',
    'how to become a radiology technician in 10 days'
]

# No instruction needed for retrieval passages
passage_prefix = ""
passages = [
    "Since you're reading this, you are probably interested in learning more about AI."
    "Below are the basic steps to becoming a radiology technician."
]

# load model with tokenizer
```

🔥 secutorpro/pocketpal-ai-landing-page

🚀 elmarcito/nvidia

🦉 geekyrakshit/medrag

🦀 ipepe/nomic-embeddings

🤖 PabloVD/CAMELSDocBot

+ 24 Spaces

📖 Collection including nvidia/NV-Embed-v2

NV-Embed 📖 Collection

NV-Embed is a ... • 3 items • Updated 19 days ago

📄 Papers for nvidia/NV-Embed-v2

NV-Retriever: Improving text embeddings

📄 Paper • 2407.15831 • Published 19 days ago • 5 likes

NV-Embed: Improved Techniques for Embedding

📄 Paper • 2405.17428 • Published 21 days ago • 21 likes

📊 Evaluation results ⓘ

accuracy ...	test set	self-reported	94.284
accuracy_s...	test set	self-reported	0.401
ap on MTE...	test set	self-reported	76.491
ap_stderr o...	test set	self-reported	1.242
f1 on MTE...	test set	self-reported	91.420
f1_stderr o...	test set	self-reported	0.504
main_scor...	test set	self-reported	94.284
accuracy ...	test set	self-reported	97.742

⌵ Expand 738 evaluations

```

model = AutoModel.from_pretrained('nvidia/NV

```



```

# get the embeddings
max_length = 32768
query_embeddings = model.encode(queries, ins
passage_embeddings = model.encode(passages, :


```



```

# normalize embeddings
query_embeddings = F.normalize(query_embeddi
passage_embeddings = F.normalize(passage_emb


```



```

# get the embeddings with DataLoader (spliti
# batch_size=2
# query_embeddings = model._do_encode(querie
# passage_embeddings = model._do_encode(pass


```



```

scores = (query_embeddings @ passage_embeddi
print(scores.tolist())
# [[87.42693328857422, 0.46283677220344543],

```

Usage (Sentence-Transformers)

```

import torch
from sentence_transformers import SentenceTr


```



```

# Each query needs to be accompanied by an c
task_name_to_instruct = {"example": "Given a


```



```

query_prefix = "Instruct: "+task_name_to_ins
queries = [
    'are judo throws allowed in wrestling?',
    'how to become a radiology technician in
]


```



```

# No instruction needed for retrieval passag
passages = [
    "Since you're reading this, you are proba
    "Below are the basic steps to becoming a
]

```

```

# load model with tokenizer
model = SentenceTransformer('nvidia/NV-Embed
model.max_seq_length = 32768
model.tokenizer.padding_side="right"

def add_eos(input_examples):
    input_examples = [input_example + model.to
    return input_examples

# get the embeddings
batch_size = 2
query_embeddings = model.encode(add_eos(quer
passage_embeddings = model.encode(add_eos(pa

scores = (query_embeddings @ passage_embeddi
print(scores.tolist())

```

License

This model should not be used for any commercial purpose. Refer the [license](#) for the detailed terms.

For commercial purpose, we recommend you to use the models of [NeMo Retriever Microservices \(NIMs\)](#).

Correspondence to

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Citation

If you find this code useful in your research, please consider citing:

```
@article{lee2024nv,  
  title={NV-Embed: Improved Techniques for T:  
  author={Lee, Chankyu and Roy, Rajarshi and  
  journal={arXiv preprint arXiv:2405.17428},  
  year={2024}  
}
```

```
@article{moreira2024nv,  
  title={NV-Retriever: Improving text embedd:  
  author={Moreira, Gabriel de Souza P and Os  
  journal={arXiv preprint arXiv:2407.15831},  
  year={2024}  
}
```

Troubleshooting

1. Instruction template for MTEB benchmarks

For MTEB sub-tasks for retrieval, STS, summarization, please use the instruction prefix template in [instructions.json](#). For classification, clustering and reranking, please use the instructions provided in Table. 7 in [NV-Embed paper](#).

2. Required Packages

If you have trouble, try installing the python packages as below

```
pip uninstall -y transformer-engine  
pip install torch==2.2.0  
pip install transformers==4.42.4  
pip install flash-attn==2.2.0  
pip install sentence-transformers==2.7.0
```

3. How to enable Multi-GPU (Note, this is the case for HuggingFace Transformers)

```
from transformers import AutoModel
from torch.nn import DataParallel

embedding_model = AutoModel.from_pretrained(
    for module_key, module in embedding_model._modules.items():
        embedding_model._modules[module_key] = DataParallel(module)
```

4. Fixing "nvidia/NV-Embed-v2 is not the path to a directory containing a file named config.json"

Switch to your local model path, and open config.json and change the value of **"_name_or_path"** and replace it with your local model path.

5. Access to model nvidia/NV-Embed-v2 is restricted. You must be authenticated to access it

Use your huggingface access token to execute *"huggingface-cli login"*.

6. How to resolve slight mismatch in Sentence transformer results.

A slight mismatch in the Sentence Transformer implementation is caused by a discrepancy in the calculation of the instruction prefix length within the Sentence Transformer package.

To fix this issue, you need to build the Sentence Transformer package from source, making the necessary modification in this line as below.

```
git clone https://github.com/UKPLab/sentence-transformers
cd sentence-transformers
git checkout v2.7-release
```

```
# Modify L353 in SentenceTransformer.py to *
pip install -e .
```